Department of Computer Science and Engineering(CSE)

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG(IIUC)

Course code:CSE-3524
Course Title:Microprocessor & Microcontroller Sessional

EXPERIMENT 3

Experiment Name: To observe the operation of arithmetic instruction.

OBJECTIVES:

1. To write a program for performing the addition of five 16 bit numbers.
2. To check the register values in kit box.
3. To analyze the result.

NECESSARY EQUIPMENT:

1. Emu8086 to run the program.
2. Notepad to write the program.
3. 8086 microprocessor kit.

PROBLEM STATEMENT:

Write an assembly language program to add five 16 bit Hex numbers. Put them in DS memory, SI= 2004H,

Exp3: Addition of five 16 bit numbers.

```
.MODEL SMALL
.DATA
.CODE

MOV SI, 2004H
MOV [SI], 6566H
MOV AX, [SI]
INC SI
INC SI
MOV [SI], 8876H
ADD  AX,  [SI]
INC SI
INC SI
MOV [SI],7755H
ADC AX,  [SI]
INC SI
INC SI
MOV [SI], 5566H
ADC AX,  [SI]
INC SI
INC SI
MOV [SI], 6677H
ADC AX,[SI]
```

DATA TABLE:-

Instruction	AX	BX	CX	DX	SI	Set Flag
MOV AX, 0010H						
MOV DS, AX						
MOV SI, 2004H						
MOV [SI], 6566H						
MOV AX, [SI]						
INC SI						
INC SI						
MOV [SI], 8876H						
ADD AX, [SI]						
INC SI						
INC SI						
MOV [SI],7755H						
ADC AX, [SI]						
INC SI						
INC SI						
MOV [SI], 5566H						
ADC AX, [SI]						
INC SI						
INC SI						
MOV [SI], 6677H						
ADC AX,[SI]						

REPORT WRITING:-

1. Include experiment name, objectives, necessary software/ equipment and problem statement.
2. Write the program with brief explanation.
3. Insert the printed copy of "list file" of the program.
4. Draw the DS memory maps.
5. Find the result of all related registers from kit box.
6. Verification of addition by theoretical calculation.
7. Write discussion if you have any error in the program or troubleshooting of the program.
8. Write the answer of the related questions given below.

RELATED QUESTIONS:-

1. Write a program to add two 32-bit numbers and store the result in DS memory.
2. What is the significance of the following instruction?

```
MOV AX, 0010H
MOV DS, AX
```

3. Why we use INC here to increment the DS's offset address? Why we have not used ADD? Discuss.

List file:

```
EMU8086 GENERATED LISTING. MACHINE CODE <- SOURCE.

noname.bin_ -- emu8086 assembler version: 4.08

[ 3/8/2025  --  2:26:34 PM ]

===============================================================
=======================================
[LINE]     LOC: MACHINE CODE                          SOURCE
===============================================================
=======================================

[   1]    0000: BE 04 20                         MOV SI, 2004H;
[   2]    0003: C7 04 66 65                      MOV [SI], 6566H;
[   3]    0007: 8B 04                            MOV AX, [SI]
[   4]    0009: 46                               INC SI
[   5]    000A: 46                               INC SI
[   6]    000B: C7 04 76 88                      MOV [SI], 8876H
[   7]    000F: 03 04                            ADD  AX,  [SI]
[   8]    0011: 46                               INC SI
[   9]    0012: 46                               INC SI
[  10]    0013: C7 04 55 77                      MOV [SI],7755H
[  11]    0017: 13 04                            ADC AX,  [SI]
[  12]    0019: 46                               INC SI
[  13]    001A: 46                               INC SI
[  14]    001B: C7 04 66 55                      MOV [SI], 5566H
[  15]    001F: 13 04                            ADC AX,  [SI]
[  16]    0021: 46                               INC SI
[  17]    0022: 46                               INC SI
[  18]    0023: C7 04 77 66                      MOV [SI], 6677H
[  19]    0027: 13 04                            ADC AX,[SI]
[  20]        :
[  21]        :
```